

BatLog v0.2 Double Antenna Instruction Manual

A Low-Cost Modular Passive Integrated Transponder (PIT) Tag Logger for Monitoring of Bat Roosting and Behavior

Purpose: This document provides standardized procedures for the assembly, testing, deployment, and maintenance of a PIT tag logging system using an RFID antenna, microcontroller, SD card data logging, and RTC timekeeping. The SOP is designed to allow replication of the system by external laboratories without prior system familiarity.

Abstract: Automated monitoring of individually marked animals enables the collection of high-resolution activity data, yet commercial systems capable of continuous individual organism tracking remain inaccessible to many research programs. BatLog is a modular, open-source passive integrated transponder (PIT) tag logging system built on Arduino-compatible hardware, costing ~3.5% of average commercial systems. BatLog is designed to extend the application of existing PIT tag infrastructure by incorporating individual identification data into continuous behavioral datasets, without requiring additional animal handling or marking. The system is well-suited for studies conducted in controlled or semi-controlled environments with defined movement points, such as nest boxes, roost entrances, burrows, or other confined passageways. BatLog is most effective in study systems where animal movement is constrained through a defined opening, and individuals reliably pass through the antenna. A dual-antenna configuration enables directional detection, allowing each tracking interaction to be classified as an animal entry or exit event. Deployed in a controlled laboratory setup at Bucknell University, USA, the BatLog system recorded 80,109 paired directional events across 27 individually identified bats. The Arduino-compatible architecture supports integration with additional sensors, including temperature, humidity, accelerometers, and acoustic monitors, making BatLog broadly applicable to behavioral and ecological studies across taxa.

Written By: Angel Yanga (Undergraduate Student, Bucknell University)

Date: December 1, 2025

Last Updated: See the most recent Git commit

Table of Contents

1. Overview	3
This SOP provides a step-by-step procedure to assemble, test, and deploy an RFID Logger system capable of reading PIT tags (134.2 kHz, FDX-B, Biomark HPT9). The system is designed to be modular, low-cost, and portable, including an integrated RTC for timestamping and microSD card data storage.	
2. Required Skills	4
2.1. Electronics Assembly & Hardware	4
2.2. Microcontroller & Embedded Systems	4
3. Materials	5
3.1. Electronics	5
3.2. Wires & Cables	5
3.3. Set Up & Mounting	7
3.4. Tools & Equipment	7
4. System Flow	8
4.1. Assemble Subassemblies	8
4.2. Assemble the Microcontroller	8
4.3. Test the System	8
4.4. Mount the System	8
4.5. Change the Power Bank Every 3-5 Days	8
4.6. Extract & Process Data	8
4.7. Interpret Results	9
5. Detailed Procedure	9
5.1. RFID Module Assembly	9
5.2. Antenna Assembly	11
5.3. RTC Module	12
5.4. 20x4 LCD Panel	13
5.5. Buck-Boost Converter	13
5.6. Microcontroller Setup (Teensy 4.1)	14
5.7. Mounting & Deployment	19
6. Troubleshooting	20
6.1. (1) SD Card not present or undetected, or (2) CSV initialization failed (or, RFID scanned but not written to CSV)	20
6.2. No RFID Scanned	21
6.3. RTC Module not found	21
6.4. LCD not found	22
7. Data Processing	22
7.1. How to extract raw data	22
7.2. How to convert raw data into Animal ID	23
7.3. Processing Data with R	24
8. Resources	24
8.1. Teensy 4.1 Code	24
8.2. R Code	24
8.3. CAD Files for 3D Prints	25
8.4. Diagrams & Photos	25
8.5. Supplementary	25

1. Overview

This SOP provides a step-by-step procedure to assemble, test, and deploy an RFID Logger system capable of reading PIT tags (134.2 kHz, FDX-B, Biomark HPT9). The system is designed to be modular, low-cost, and portable, including an integrated RTC for timestamping and microSD card data storage.

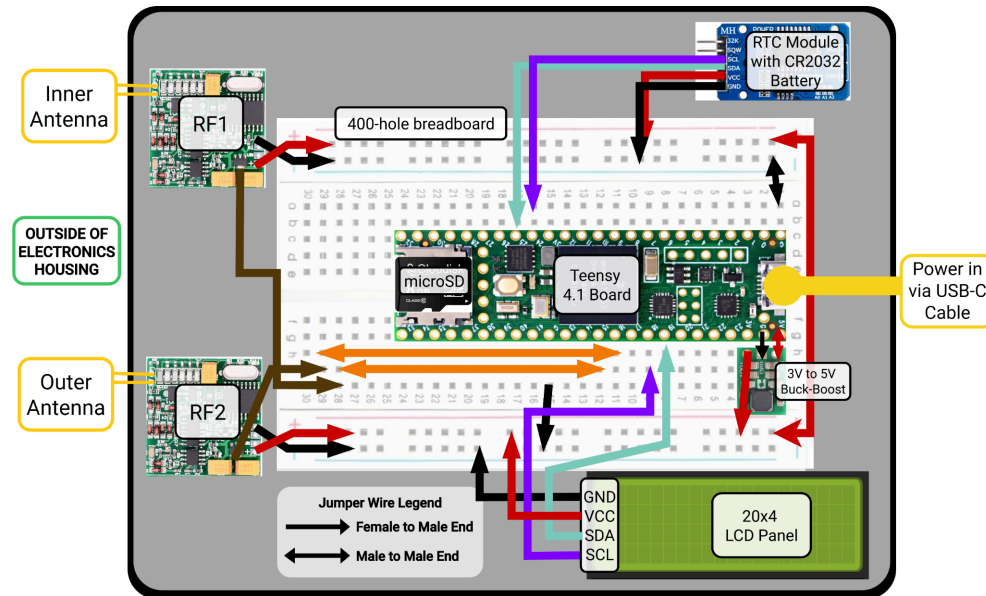


Figure 1. Electronics schematic for the two-antenna PIT tag logging system, showing the Teensy 4.1 microcontroller, double RFID-antenna subsystem, RTC module, 20x4 LCD Panel with Backpack, and power distribution for bat box data collection.

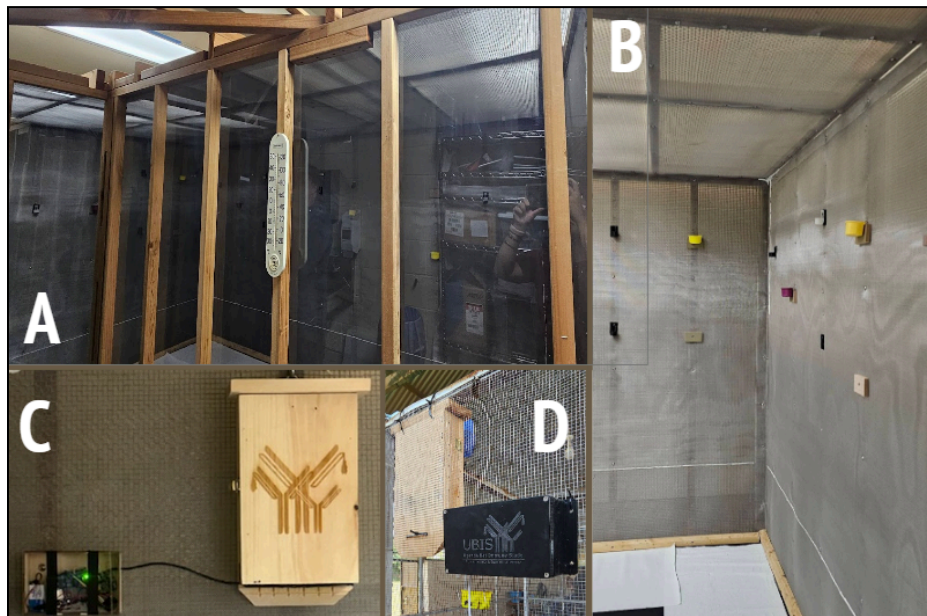


Figure 2. Overview of the flight cage and experimental setups. (A) Exterior view of the flight cage at Bucknell University, showing the acrylic glass panels. (B) Interior view of

the flight cage, highlighting the mesh structure. (C) Experimental setup at Bucknell University, where the system was mounted using wall anchors. (D) Field setup in Uganda, where the wooden bat box was placed inside the enclosure, and the Batlog electronics were mounted externally and secured to the mesh using zip ties.

2. Required Skills

2.1. Electronics Assembly & Hardware

1. Through-hole Soldering

- Installing pin headers on PCBs
- Identifying a functional solder joint
- How to solder: [Soldering Tutorial for Beginners: Five Easy Steps](#)

2. Crimping Dupont Connectors

- Properly crimping thin wires to attach dupont connectors
- Using a ratcheting crimper correctly
- How to crimp for dupont connectors:
[Dupont Connectors - Quickly and easily make your own](#)

3. Basic Wiring Practices

- Correct power, ground, and signal routing
- Follow color-coded wire management
- Know the differences and function of different terminal on PCBs (microcontrollers and modules)
- Tips for working with dupont wires:
[#12 Five Tricks for working with Dupont wires](#)

4. Multimeter Use

- Measuring voltage and current to verify functional connections
- It's helpful to measure voltage for power connections
- It is more useful to measure current between non-power connections (e.g. TX pin to RX pin)
- How to use a multimeter:
[How to Use a Multimeter for Beginners - How to Measure Voltage,...](#)

2.2. Microcontroller & Embedded Systems

1. Arduino IDE Proficiency

- Selecting the correct board and port
- Reliably upload firmware to the system
- How to install and run Arduino IDE:
[How to Install Arduino Software IDE on Computer / Laptop](#)

2. R Studio Proficiency

- Effectively import the raw data, metadata, and produce effective visualizations of findings
- How to install and run R Studio:
[How to install R and RStudio on Windows 11 \(Updated 2025\)](#)

3. Basic Embedded Debugging

- a. Interpreting Serial Monitor output
- b. Identifying whether failures are firmware vs. hardware

4. RTC Handling

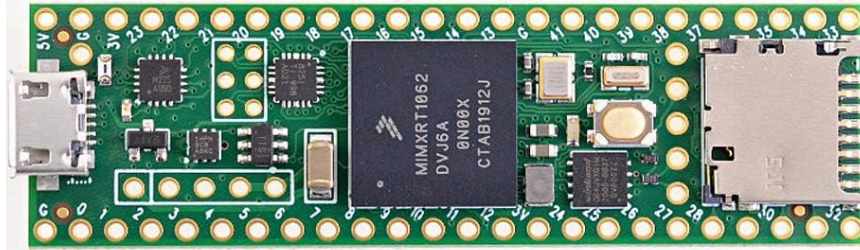
- a. Flashing code properly to initialize RTC time
- b. Know how to replace the coin battery without losing the stored time
 - i. You plug in the system to a reliable power source through the Arduino R4's Type C input.
 - ii. Then, you replace the coin battery

3. Materials

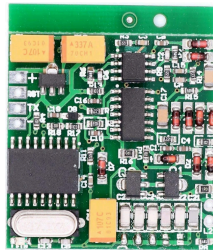
See the full bill of materials in [Google Drive](#) for links to materials.

3.1. Electronics

- 1x PJRC Teensy 4.1 ARM Cortex-M7 Processor at 600MHz with a NXP iMXRT1062



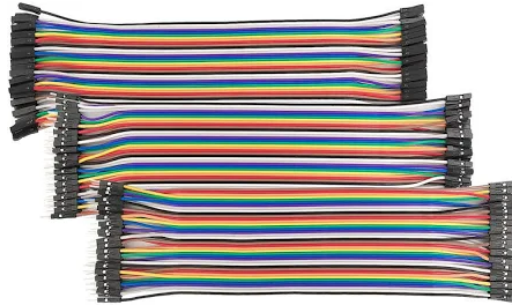
- 1x RFID Module (Reads 134.2 kHz, FDX-B)



- 1x DS3231 Real Time Clock Module
- 1x 5V, 1A Step-Up/Step-Down Voltage Regulator (S13V10F5)
- 1x IIC I2C 2004 LCD Module 20x04 with Backpack
- 1x 2GB microSD Card (Cloudisk)
- 1x 6 1/8 in x 1.375 in x 1/4 in 3D-printed antenna frame
- 1x Roll 24 AWG Copper Wire Antenna
- 1x CR2032 Li-ion Coin Battery

3.2. Wires & Cables

- 1x microUSB to USB A Cable, 10 ft, nylon braided
- 1x Dupont Jumper Wire Kit (40 male-to-male, 40 female-to-female, 40 male-to-female, 10 cm, multicolor)



- 1x Dupont Connector Kit, unsoldered, to assemble module connections
 - Needs to include male and female dupont connectors (aluminum part) and headers (plastic part)



- 1x Unsoldered Pin Header Kit
 - Could already be included in dupont connector kit, but not required



- 1x Jumper Wire Kit Solderless Flexible



3.3. Set Up & Mounting

- 1x Electronics Box; water-resistant, ABS Plastic, includes lid (7.87"x4.72"x2.95")



- 1x Anker Power Bank (PowerCore 10K, 5V/3A USB-C, 10,000 mAh)
- 1x Plastic container without lid
 - Should fit the 5.99 in × 2.81 in × 0.61 in power bank when turned over
- 4x Nylon Zipties (2.5x150 mm)
- 1x Wooden Bat Roost Box
 - *Ours was modified from a commercial bat box kit*

3.4. Tools & Equipment

- 1x Soldering Station with Accessories (Lead, flux, steel wool, lead sucker)

Soldering Station



Accessories



- 1x Wire Stripper & Cutter
- 1x Philips Screwdriver
- 1x Multimeter
- 1x Wire Crimper (0.25-size slot)
- 1x Roll Electrical Tape
- 1x Roll Double-Sided Adhesive

Reference: Uline Removable Double-Sided Foam Squares - 1 x 1", 1/16" Thick, but anything similar works.



- 1x MicroSD card reader to USB A/C
- 1x Computer with Arduino IDE and R
- 1x Biomark Test Tag 12mm FDX-B Key Chain



4. System Flow

4.1. Assemble Subassemblies

The following subassemblies may be prepared simultaneously by different personnel to reduce the total build time:

- [Section 4.1] RFID Modules
- [Section 4.2] Antennas
- [Section 4.3] RTC Module
- [Section 4.4] Buck-Boost Converter
- [Section 4.5] 20x4 LCD Screen

4.2. Assemble the Microcontroller

This will be the main assembly of the system, where you put all of your subassemblies together.

4.3. Test the System

Verify that the assembled microcontroller and all connected subassemblies (RFID module, antenna, SD card module, and RTC battery holder) power on correctly, communicate as expected, and record data without errors.

4.4. Mount the System

Securely install the fully assembled system at the designated deployment location, ensuring all components are properly fastened, protected, and oriented for reliable operation.

4.5. Change the Power Bank Every 3-5 Days

Replace the power bank at regular 3-5 day intervals to maintain uninterrupted power and continuous data collection.

4.6. Extract & Process Data

Retrieve the recorded data from the system and clean, organize, and process it for analysis.

4.7. Interpret Results

Analyze the processed data to identify patterns, trends, and findings relevant to the study objectives.

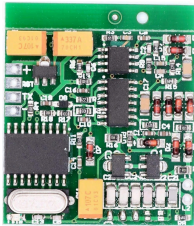
5. Detailed Procedure

5.1. RFID Module Assembly

The following procedure is written for ONE RFID Module. For this system, you will need TWO RFID Modules, so you must repeat this procedure twice.

Materials & Tools

1. 1x RFID Module (Reads 134.2 kHz, FDX-B)



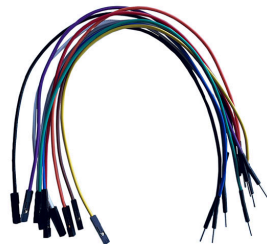
2. 1x 1×4 unsoldered pins
 - a. Cut the long strip of unsoldered pin headers into 1x4:



3. 1x 1×2 unsoldered pins
 - a. Cut the long strip of unsoldered pin headers into 1x2:

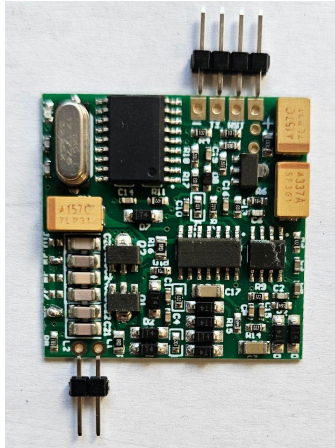


4. 5x Female to male dupont jumper wires
 - a. Red, Black, Brown, Purple, Gray



5. 1x Soldering Station with Accessories

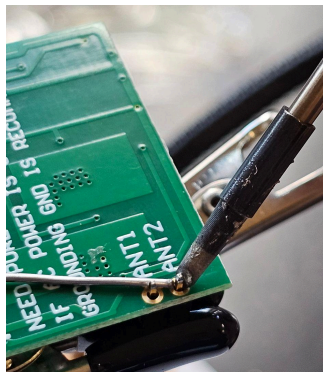
Procedure



1. Heat the soldering iron to $\sim 310\text{-}320^{\circ}\text{C}$.
 - a. You can heat it to 380°C if needed, as some soldering leads melt at a higher temperature.
2. Insert the dupont 1 $\times$ 2 header into the antenna input holes using the same orientation.

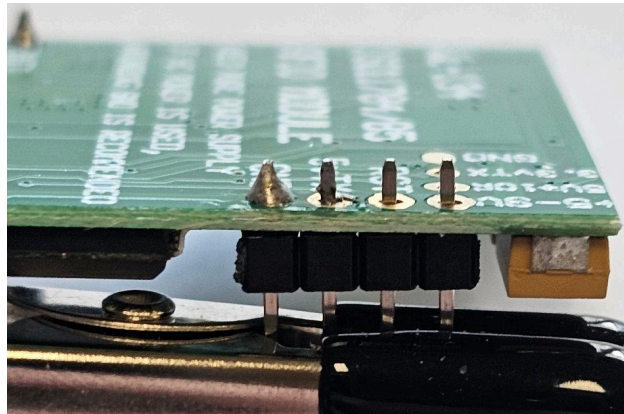


3. Solder both antenna pins.



4. Long pins on top, short pins on bottom.
 - a. The long side of the headers will be where your wires can connect, so

- make sure you leave them in the most accessible spot for future wiring
5. Solder the two pins securely.
 6. Insert the dupont 1×4 header into the RFID module output holes from the back.
 - a. Output holes will be recognized by:
 - i. TX
 - ii. RX (won't be used, but it's convenient to solder)
 - iii. 5-9+V (or 3.3V)
 - iv. GND



7. Solder all four pins securely.
8. Attach female dupont wires using the following color convention:
 - a. +5–9 V → Red
 - b. TX → Brown
 - c. GND → Black
 - d. ANT1 → Purple
 - e. ANT2 → Gray

5.2. Antenna Assembly

The following procedure is written for ONE Antenna. For this system, you will need TWO Antennas, so you must repeat this procedure twice.

Materials & Tools

1. 1x 6 1/8 in x 1.375 in x 1/4 in 3D-printed antenna frame
2. 1x Roll 24 AWG Copper Wire Antenna
3. 1x Roll Electrical Tape

Procedure

1. Wrap the 24 AWG copper wire on one side of your mold with about 5 inches of spare sticking out
 - a. You need the space copper wire to connect to an RFID Module, so this is very important
2. Wrap the rest of the 24 AWG wire around the antenna mold approximately 40 turns.
 - a. Avoid crossing turns excessively to make sure it is as flat as possible after each wrap.

3. At the end, leave the same length of copper wire sticking out as in step 1.
 - a. This is your second antenna lead to connect to the RFID Module
4. Carefully use electrical tape to cover exposed wires.
5. Progress outward slowly, ensuring the wire does not unravel.
6. Fully secure the antenna with electrical tape once wrapping is complete.

Notes

- I do not have sufficient images for this process step-by-step. So, you can use this video (<https://youtu.be/IRoxIW0Hx-g>: 8:37-12:40) as a visualization.
 - After wrapping the antenna around the PVC, we would wrap our antenna with electrical tape. This is for protection for and form the bats.
- This was the most sensitive step for our production of 10 BatLogs for the Moco 2025 field study. This is due to the size of the copper wire we used. It was too thin to crimp, so we needed to get each crimp perfectly.
 - If you plan on tracking larger animals, thicker copper wire would be ideal and you might not have this problem.
- If the crimping is giving you a difficult time, a transition to soldered JST connectors could be helpful.
 - We soldered the antenna leads (the excess antenna wire) to a male JST connector. Then, the antenna leads on the RFID modules were soldered with a female-ended JST connector.
 - The difficult part here would be soldering very tiny parts together, but this method was more secure than crimping.

5.3. RTC Module

Materials and Tools

1. 1x DS3231 Real Time Clock Module
2. 1x 1×6 unsoldered pins



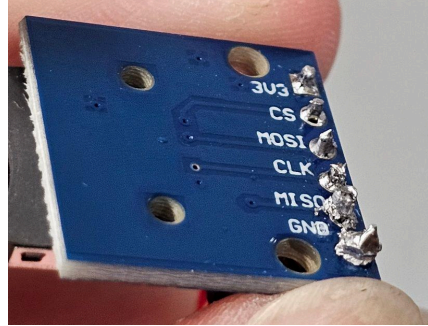
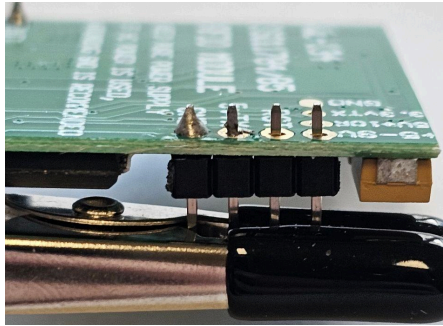
3. 4x Female to Male dupont jumper wires
 - a. Purple, Gray, Red, Black
4. 1x 2CR2032 Battery
5. 1x Soldering Station with Accessories

Procedure

** If your RTC Module comes with pre-soldered pins, you can skip steps 2 through 5*

1. Heat the soldering iron to ~310-320°C.
 - a. You can heat it to 380°C if needed, as some soldering leads have a higher melting point.
2. Insert the 1×6 unsoldered pins into the RTC module from the back side of the PCB.

- a. The long pins must extend through the top of the module.
 - b. The short pins remain on the back side of the PCB.
3. Verify correct orientation before soldering (see reference image).



- a. Heat both the pin and the pad simultaneously before applying solder.
 - b. Ensure the solder fully wets both surfaces with no visible gaps between the pin and the pad.
 - c. A proper solder joint should resemble an inverted cone extending from the pad to the pin.
4. Solder each header pin to its corresponding contact hole on the RTC module.
5. Visually inspect all solder joints and gently tug each pin to confirm stability.
6. Attach female dupont jumper wires to the header pins using the following color-to-pin mapping:
 - a. SCL → Purple
 - b. SDA → Gray
 - c. VCC → Red
 - d. GND → Black
7. Insert the CR2032 Battery into the coin battery slot of the RTC module.
8. Ensure the battery is oriented correctly and fully seated in the module's slot according to its orientation.

5.4. 20x4 LCD Panel

Materials and Tools

1. 1x IIC I2C 2004 LCD Module 20x04 with Backpack
2. 4x Female to Male dupont jumper wires
 - a. Purple, Gray, Red, Black

Procedure

** Your LCD panel backpack should come in pre-soldered with the panel itself*

1. Attach female dupont jumper wires to the header pins using the following color-to-pin mapping:
 - a. SCL → Purple
 - b. SDA → Gray
 - c. VCC → Red
 - d. GND → Black

5.5. Buck-Boost Converter

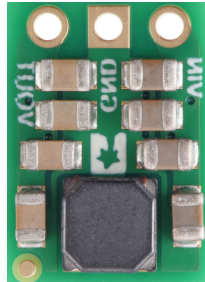
Materials

1. 1x 5V, 1A Step-Up/Step-Down Voltage Regulator (S13V10F5)
2. 2x 1x1 dupont header

Procedure

1. Solder the 1x1 dupont header with the long side poking above the front of buck-boost converter (step-up/down voltage regulator)

Front:



2. Solder the 1x1 dupont header with the long side poking below the front of the buck-boost converter
 - a. This is what goes into the breadboard, while the other pokes above it for easy dupont wire jumper attachment in later assembly

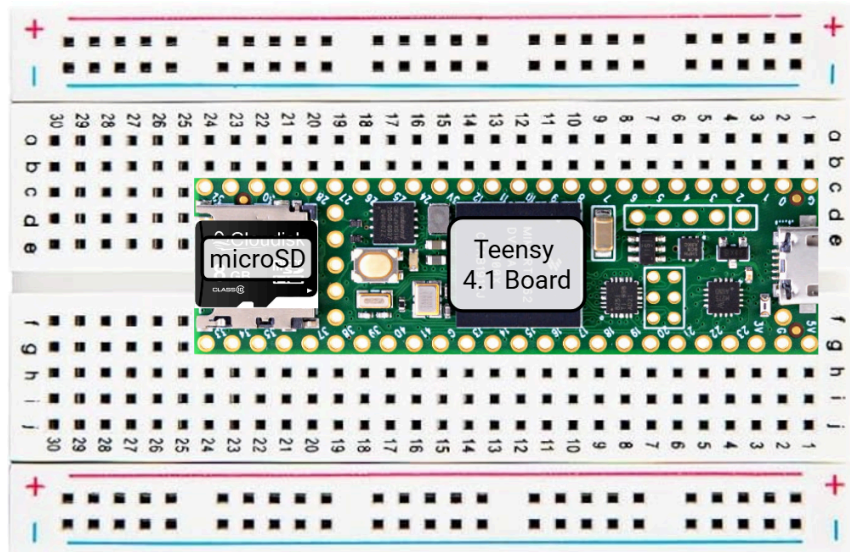
5.6. Microcontroller Setup (Teensy 4.1)

Materials

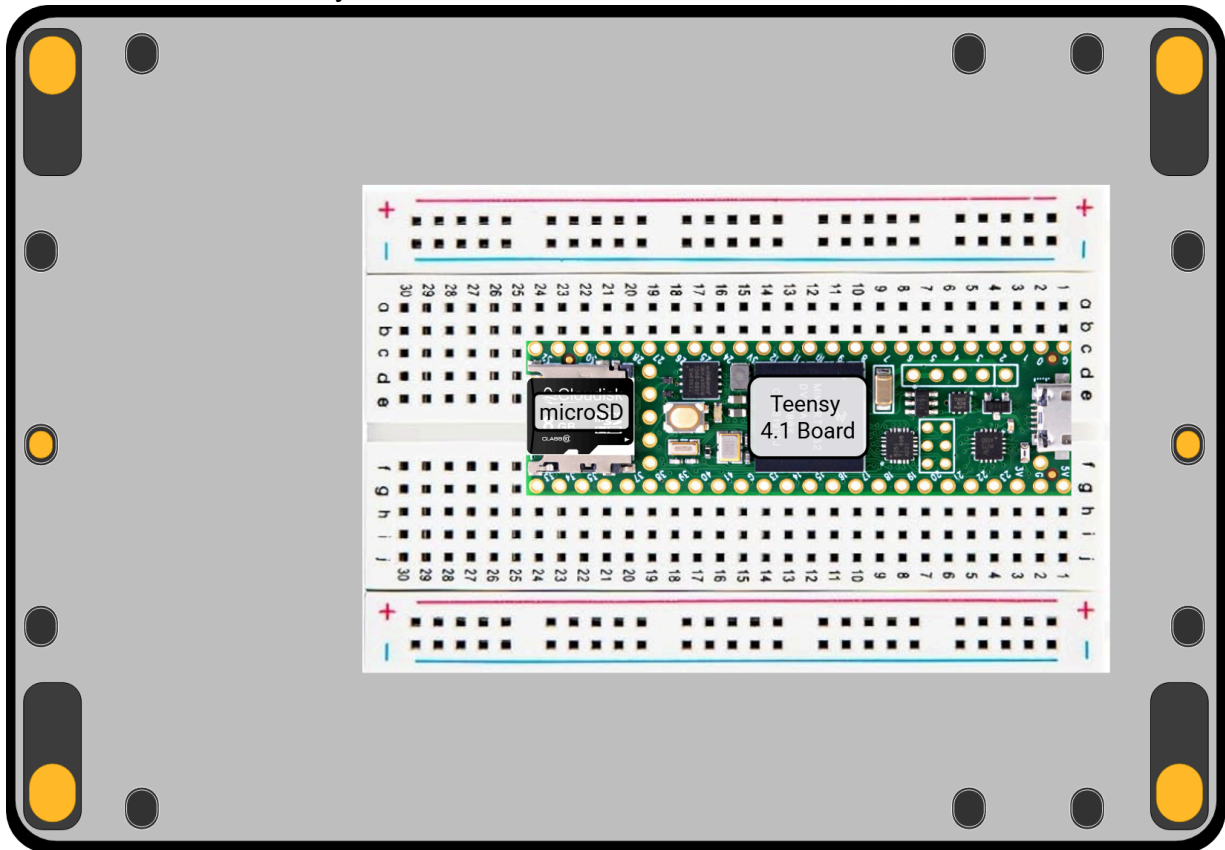
1. 1x Teensy 4.1
2. 2x RFID Module Subassembly
3. 2x Antenna Subassembly
4. 1x RTC Module Subassembly
5. 1x 20x4 LCD Panel Subassembly
6. 1x Electronics Box; water-resistant, ABS Plastic, includes lid (7.87"x4.72"x2.95")
7. 1x microUSB to USB A Cable, 10 ft, nylon braided
8. 1x Computer with Arduino IDE and Arduino Script
9. 1x Roll Double-Sided Adhesive

Procedure

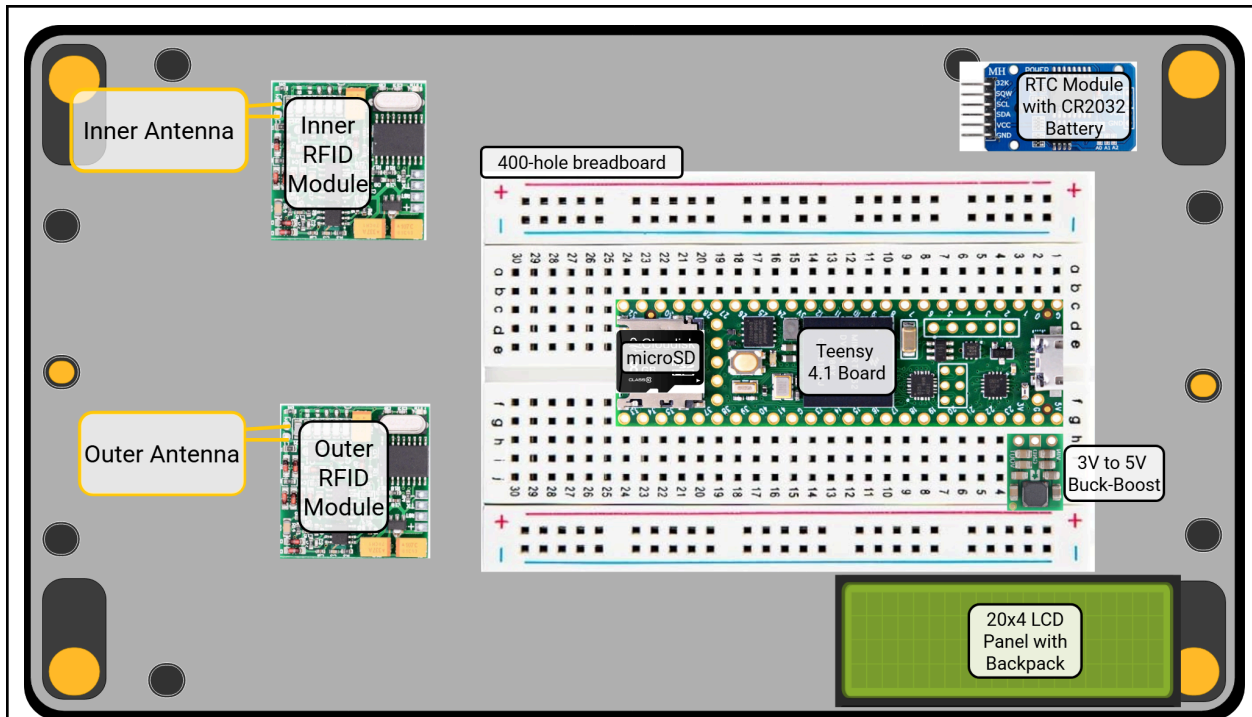
1. The Teensy 4.1 is ready for assembly right out of the box.
 - a. It is recommended to test the board with pre-installed test scripts on the Teensy 4.1
2. Pin down the Teensy 4.1 board onto the 400-hole breadboard. Then, insert the microSD card in the Teensy 4.1 slot.



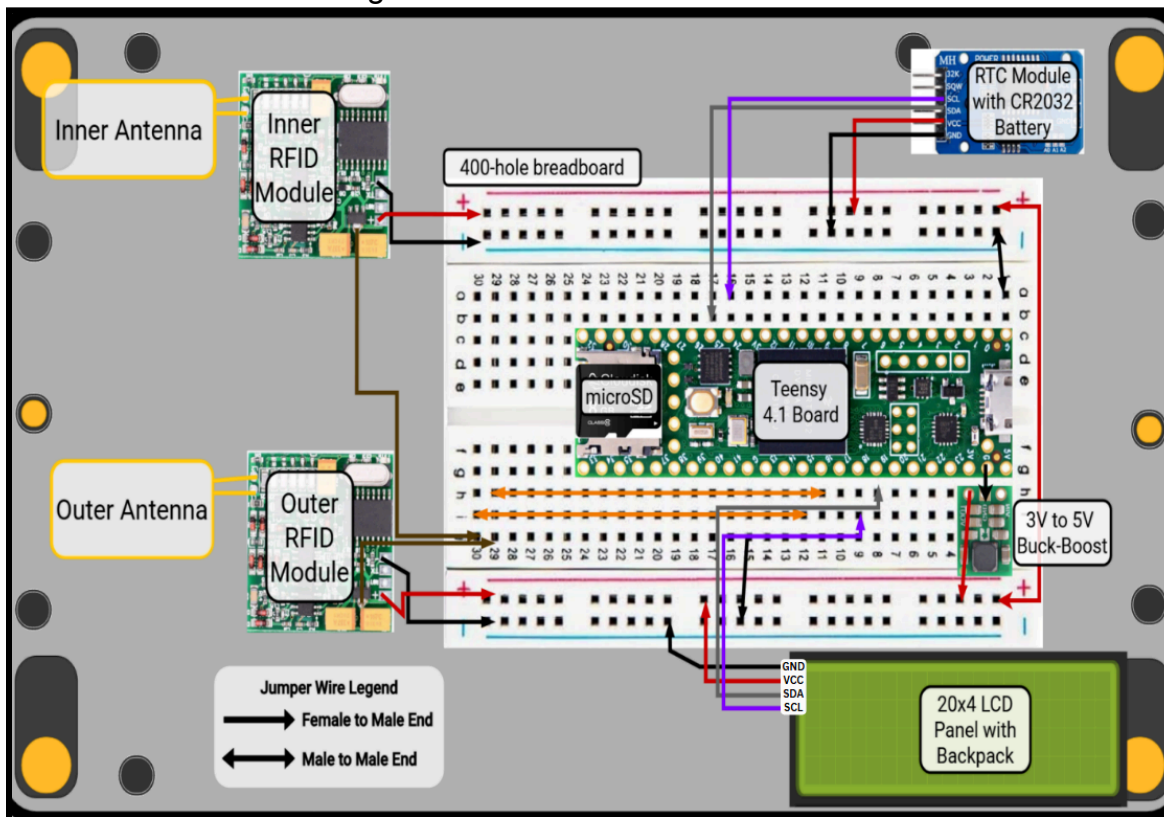
3. Mount the Teensy 4.1 in the electronics box with double-sided adhesive.



4. Mount the remaining subassemblies using double-sided adhesive, as shown in the image, except for the antenna.



5. Use all of the dupont wires from the subassemblies to connect to the Arduino
 - a. Follow the diagram below:



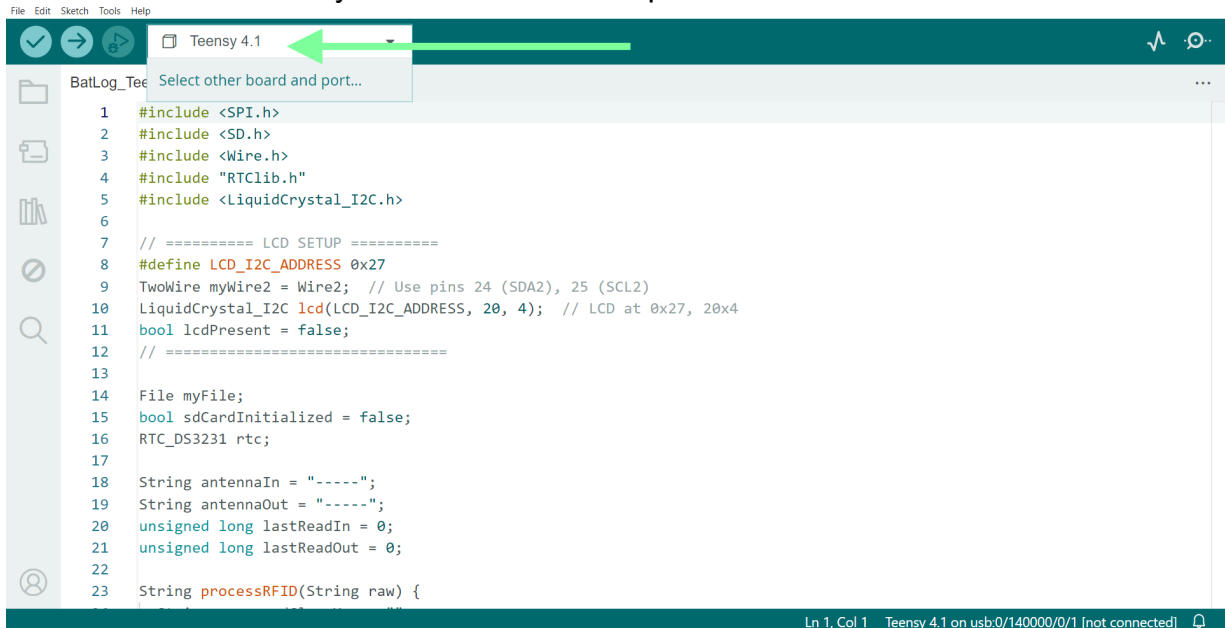
6. Connect the antenna with the gray and purple wires of the RFID Module
7. Connect the system to your computer with the Arduino IDE with the charging

cable

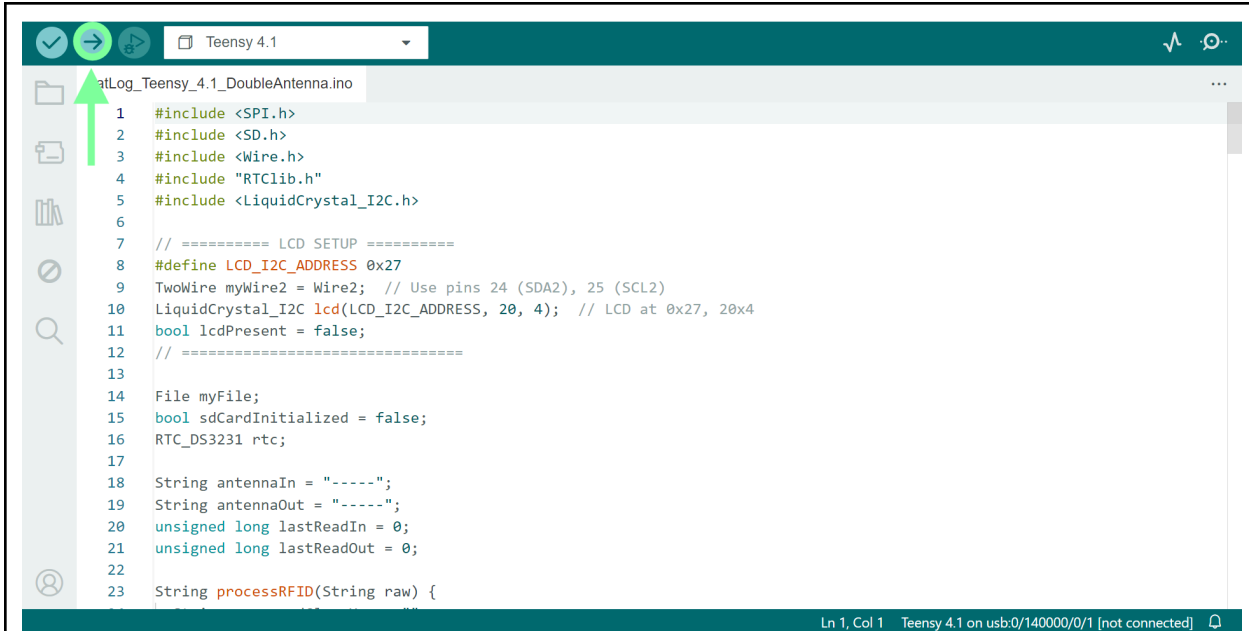
- a. A cable hole through your electronics box is necessary, so you can have the charging cable run through it once the lid is closed
8. Set up your Arduino IDE with Teensy 4.1 drivers
 - a. See more detailed procedure online
 - i. [PJRC \(Teensy Manufacturer\) Software Setup Tutorial with Arduino IDE](#)
 - ii. [How to Program Teensy with Arduino IDE \(Maker Tutor via YouTube\)](#)
9. Open the Arduino Script file (ino file) on Arduino IDE ("BatLog_Teensy_4.1_DoubleAntenna.ino")

Name	Type	Size
 BatLog_Teensy_4.1_DoubleAntenna.ino	INO File	7 KB

10. Select the Teensy 4.1 on the board dropdown



11. Upload the code onto the Teensy 4.1 by clicking the right-pointed arrow on the upper left corner of the IDE



```

1  #include <SPI.h>
2  #include <SD.h>
3  #include <Wire.h>
4  #include "RTCLib.h"
5  #include <LiquidCrystal_I2C.h>
6
7  // ===== LCD SETUP =====
8  #define LCD_I2C_ADDRESS 0x27
9  TwoWire myWire2 = Wire2; // Use pins 24 (SDA2), 25 (SCL2)
10 LiquidCrystal_I2C lcd(LCD_I2C_ADDRESS, 20, 4); // LCD at 0x27, 20x4
11 bool lcdPresent = false;
12 // =====
13
14 File myFile;
15 bool sdCardInitialized = false;
16 RTC_DS3231 rtc;
17
18 String antennaIn = "-----";
19 String antennaOut = "-----";
20 unsigned long lastReadIn = 0;
21 unsigned long lastReadOut = 0;
22
23 String processRFID(String raw) {

```

12. Open the Serial Monitor on Arduino IDE (Ctrl+Shift+M)

13. Press the white reset button on the Teensy 4.1

14. You will see messages on the Serial Monitor

a. If your wiring is functional, you should see this message:

```

Starting Serial Communication...
RFID #1 Serial Ready (Pin 16)...
RFID #2 Serial Ready (Pin 15)...
Initializing Onboard SD card...Onboard SD card initialized.
RFID_Log.csv opened and header checked.
Starting RTC Module and Setting Time
Ready to read RFID from both readers...

```

15. Put the Biomark HPT9 Tag tester through the antenna

a. When your system is functional, here is the output on the Serial Monitor:

```

Timestamp: 12/28/2025-16:45:11, Reader: RFID #2, Raw RFID:
0FA02E7700DD30010000000000r, Processed RFID: 02640
Data written to Onboard SD card.

```

b. Here are example outputs when your system has issues:

i. SD Card not present or undetected

```

Initializing Onboard SD card...
Card failed or not present. RFID data will still be displayed on Serial Monitor.

```

ii. No RFID Scanned

```

Starting Serial Communication...
RFID #1 Serial Ready (Pin 16)...
RFID #2 Serial Ready (Pin 15)...
Initializing Onboard SD card...Onboard SD card initialized.
RFID_Log.csv opened and header checked.
Starting RTC Module and Setting Time
Ready to read RFID from both readers...

```

(no RFID reads are coming in the serial monitor)

iii. RTC Module not found

Couldn't find RTC on Wire2. Time will not be logged.

iv. LCD not found

LCD not found on Wire.

v. CSV initialization failed

SD card not initialized. Data not logged to SD.
Error opening RFID_Log.csv for writing. Data not logged to SD.

16. You are ready to mount the system once it is functional. If not, go through the troubleshooting section.

17. Once you have troubleshooted your errors, you can mount your system.

5.7. Mounting & Deployment

Materials

1. 1x Electronics Box; water-resistant, ABS Plastic, includes lid (7.87"x4.72"x2.95")
 - a. 4x included Philips screws for the lid
2. System electronics (fully assembled and tested)
3. 4x Nylon Zipties (2.5x150 mm)
4. 1x Wooden Bat Roost Box
5. 1x Anker Power Bank (PowerCore 10K, 5V/3A USB-C, 10,000 mAh)
6. 1x microUSB to USB A Cable, 10 ft, nylon braided
7. 1x Plastic container without lid
 - a. Should fit the 5.99 in × 2.81 in × 0.61 in power bank when turned over
8. 1x 2GB microSD Card (Cloudisk)
9. 1x MicroSD card reader to USB A/C
10. 1x Computer for file backup

Procedure

1. Unplug the system from the computer.
2. Unplug the antenna from the system. It will be reattached once mounting is done.
3. Secure the enclosure lid onto the electronics box using four screws, ensuring the charging cable remains routed through the designated opening in the box.
4. Align the electronics box behind the wooden roost box.
5. Use zip ties to mount the electronics box to the outside of the flight cage, positioned near or directly behind the wooden roost box.
 - a. The system must be close enough to the roost box opening for the antenna to reach it from behind.
6. Open the wooden bat roost box.
7. Mount the inner antenna about 2 inches inside the box.
 - a. Use electrical tape to secure the antenna.
 - b. Leave the two antenna leads sticking out through the entrance of the

- box. These leads must be able to pass through the cage mesh and connect to the RFID module mounted outside the cage.
8. Mount the outer antenna directly at the entrance of the wooden roost box.
 - a. Use electrical tape to secure the antenna.
 - b. Make sure there are no gaps between the wooden entrance and the antenna so the bats cannot pull it off.
 9. Connect the antenna leads to the RFID modules.
 - a. If needed, poke holes in the cage mesh to allow the DuPont leads from both antennas to pass through to the electronics system outside the cage.
 - b. The inner antenna connects to the top (inner) RFID module, and the outer antenna connects to the bottom (outer) RFID module. Follow the image below to ensure proper connections.
 10. Connect the system to the power bank using the USB Type-C cable.
 11. Place the power bank on top of the flight cage and cover it with a plastic container to protect it from water exposure.
 12. Replace the power bank every 3–5 days.
 - a. Although the system can theoretically operate for up to 5 days, field testing showed that replacing the power bank every 3 days leaves approximately 20% charge remaining, providing a safety margin.
 13. To check the collected data:
 - a. Unscrew the enclosure lid.
 - b. Remove the microSD card from the slot.
 - c. Insert the microSD card into a computer and open the CSV file.
 14. Verify that the CSV file contains accurate timestamps and valid PIT tag readings.
 - a. If both are correct, the system is functioning properly.
 15. Reinsert the microSD card into the system and reseal the enclosure lid.
 16. Do not re-upload the firmware if the system is functioning correctly.
 - a. After reinserting the microSD card, simply press the reset button on the microcontroller and leave the system running.
 17. Once you have completed the experiment, perform the data extraction process (Section 7) before unmounting the system for cleanup.
 18. You must cut the zipties off the mesh wall to unmount

Notes

- I do not have sufficient photos of the mounting process. I attached images in the Google Folder ([link](#)) for your reference to see how the system was mounted.

6. Troubleshooting

- 6.1. (1) SD Card not present or undetected, or (2) CSV initialization failed (or, RFID scanned but not written to CSV)

Sample Serial Output

Output 1	Output 2
----------	----------

Initializing Onboard SD card...
Card failed or not present. RFID data
will still be displayed on Serial
Monitor.

SD card not initialized. Data not logged
to SD.
Error opening RFID_Log.csv for writing.
Data not logged to SD.

Potential Issues & Fixes

1. The SD Card wires are wired incorrectly
 - a. Try to rewire the jumper cables between the SD card subassembly and Arduino
 - b. Unplug all the wires on the Teensy 4.1 from the SD Card subassembly
 - c. Unplug the wires on the SD Card Assembly
 - d. Follow Procedure Step 5 from Section 5.6 Microcontroller Setup (Teensy 4.1)
 - e. Re-run the system and try to read your tester again
2. The wiring is functional, but the SD card is not being read
 - a. Take out the microSD card by sliding it out
 - b. Wipe the microSD card leads with a microfiber cloth
 - c. Insert the microSD card into the module securely
 - d. Re-run the system and try to read your tester again
3. The wiring is functional, but the SD card isn't all the way in
 - a. Remove and reinstall the microSD card
 - b. Re-run the system and try to read your tester again

6.2. No RFID Scanned

Sample Serial Output

Starting Serial Communication...
Ready to read RFID...
Starting RTC Module and Setting Time

Potential Issues & Fixes

1. The antenna is not connected properly
 - a. Unplug and replug the dupont connections
2. The antenna dupont heads are not properly crimped
 - a. You can re-crimp the ends again to secure the female head to it properly
3. The antenna is not coiled sufficiently
 - a. You need to replace another antenna
 - b. You need to maintain a strong tight connection between each wire
 - c. You need to make sure you wrap it thick enough. You can loop the wire up to 60-70 times.
4. The antenna coil loop is broken
 - a. The coil loop can be broken with rigid corners, eventually cutting the wires in half
 - b. You need to replace another antenna

6.3. RTC Module not found

Sample Serial Output

Couldn't find RTC on Wire2. Time will not be logged.

Potential Issues & Fixes

1. Wiring wasn't done properly
 - a. You need to rewire the system, similar to Section 6.1 Potential Issues & Fixes 1. Instead, follow Section 5.3 under Procedure Step 6

Notes

- This won't be a coin battery issue because the system will have power through the Teensy 4.1.
 - A dead coin battery could only result in a reset of date and time to 0/0/0-0:00:00.
 - When this happens, you need to replace the battery and re-upload the code into the Teensy 4.1, so you can get the new time for the RTC module to save

6.4. LCD not found

Sample Serial Output

LCD not found on Wire.

Potential Issues & Fixes

1. Wiring wasn't done properly
 - a. You need to rewire the system, similar to Section 6.1 Potential Issues & Fixes 1. Instead, follow Section 5.4 under Procedure Step 1

Notes

- You don't need this LCD panel for the system to work properly
- This was added to the system so that you can easily see how the system was doing. The Teensy 4.1 code will operate fine without it, so long as the Serial Monitor printouts are functional.

7. Data Processing

7.1. How to extract raw data

Materials

1. 1x Electronics system
2. 1x MicroSD card reader to USB A/C
3. 1x Philips Screwdriver
4. 1x Computer for file backup
 - a. You can also use your phone in the field

Procedure

1. Unscrew the Phillips screws that secure the lid.
2. Remove the lid of the system.
3. Remove the SD card from the module by sliding it out.
4. Insert the SD card into the card reader.
5. Plug the card reader into your computer.
6. Save the CSV file containing the raw data locally for later data processing.

7.2. How to convert raw data into Animal ID

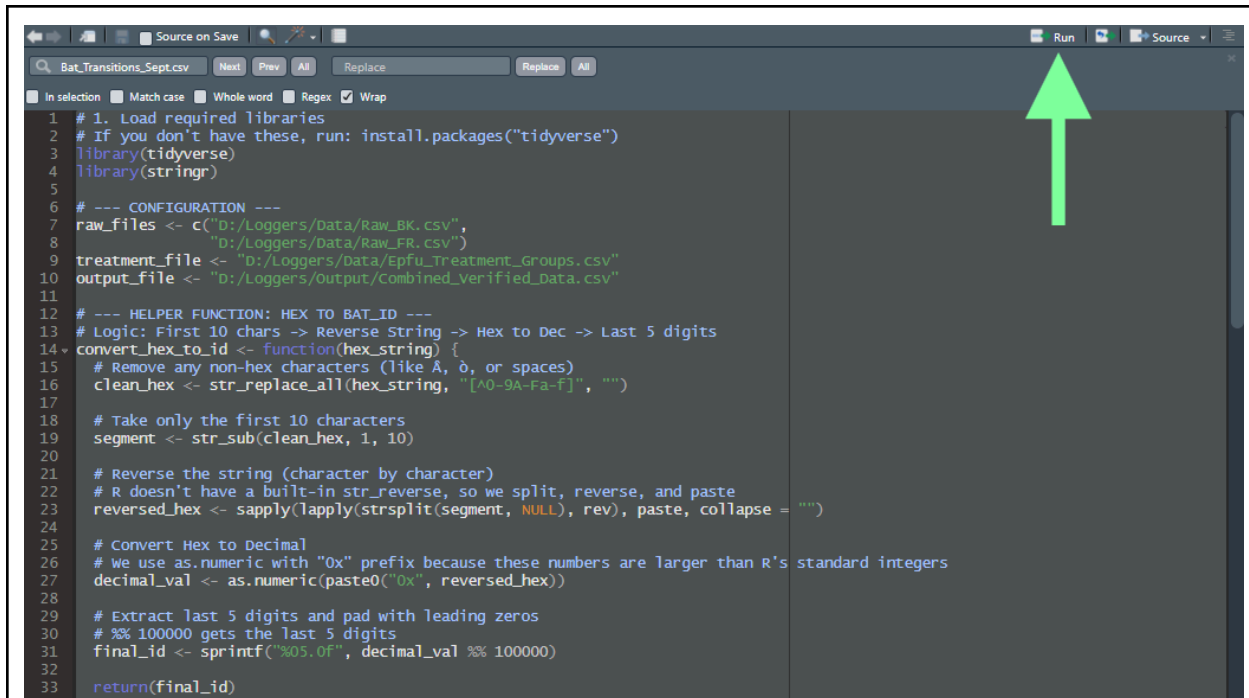
Our RFID module reads input from the RFID tags in reversed decimal (base-12) order. In the lab, animal IDs are defined as a combination of the project leader's initials and the last five digits of the PIT tag's hexadecimal code. As a result, the raw PIT tag data must be further processed through R.

Procedure

1. Download "RawData-ConversiontoBatID.R" from Google Drive Link
2. Create a folder for your new R script
3. Open the RawData-ConversiontoBatID.R script in R Studio
4. Place your raw CSV files in the same folder as the Conversion Script
 - a. You might have to rename the raw CSV files to distinguish the different places they were pulled from.
 - b. Then, you need to change the file names/directories manually in the code
 - i. Line 7-8: Raw CSV files location
 - ii. Line 9: Location for treatment groups and other metadata for the animals
 1. You don't use this information yet. This will be used in the next step
 - iii. Line 10: Location and name for your output CSV file

```
1 # 1. Load required libraries
2 # If you don't have these, run: install.packages("tidyverse")
3 library(tidyverse)
4 library(stringr)
5
6 # --- CONFIGURATION ---
7 raw_files <- c("D:/Loggers/Data/Raw_BK.csv",
8               "D:/Loggers/Data/Raw_FR.csv")
9 treatment_file <- "D:/Loggers/Data/Epfu_Treatment_Groups.csv"
10 output_file <- "D:/Loggers/Output/Combined_Verified_Data.csv"
```

5. Select all the lines of code and click Run



```

1 # 1. Load required libraries
2 # If you don't have these, run: install.packages("tidyverse")
3 library(tidyverse)
4 library(stringr)
5
6 # --- CONFIGURATION ---
7 raw_files <- c("D:/Loggers/Data/Raw_BK.csv",
8               "D:/Loggers/Data/Raw_FR.csv")
9 treatment_file <- "D:/Loggers/Data/EpFu_Treatment_Groups.csv"
10 output_file <- "D:/Loggers/Output/Combined_Verified_Data.csv"
11
12 # --- HELPER FUNCTION: HEX TO BAT_ID ---
13 # Logic: First 10 chars -> Reverse String -> Hex to Dec -> Last 5 digits
14 convert_hex_to_id <- function(hex_string) {
15   # Remove any non-hex characters (like A, b, or spaces)
16   clean_hex <- str_replace_all(hex_string, "[^0-9A-Fa-f]", "")
17
18   # Take only the first 10 characters
19   segment <- str_sub(clean_hex, 1, 10)
20
21   # Reverse the string (character by character)
22   # R doesn't have a built-in str_reverse, so we split, reverse, and paste
23   reversed_hex <- sapply(strsplit(segment, NULL), rev, paste, collapse = "")
24
25   # Convert Hex to Decimal
26   # We use as.numeric with "0x" prefix because these numbers are larger than R's standard integers
27   decimal_val <- as.numeric(paste0("0x", reversed_hex))
28
29   # Extract last 5 digits and pad with leading zeros
30   # %% 100000 gets the last 5 digits
31   final_id <- sprintf("%05.0f", decimal_val %% 100000)
32
33   return(final_id)

```

6. You should be able to use the newly made CSV file for data processing

7.3. Processing Data with R

Procedure

1. Download “Data-Processing-and-Visualization.Rmd” from Google Drive Link
2. Create a folder for your new R script
3. Copy your output CSV from Section 7.2 into the same folder as the previous step in this section
4. Download and place the metadata for your study into the same folder
5. Move the files downloaded in Steps 2 and 3 into the new project folder created in Step 1.
6. Open the “Data-Processing-and-Visualization.Rmd” file in R Studio
7. Run the code blocks from lines 8 through 110 to process the entries, exits, and lingers of the bats
8. Starting at line 112, you can customize how you want to use the processed datasets.
 - a. You can see different versions of data visualizations presented

8. Resources

File resources are attached in the shared BatLog Google Drive folder

8.1. Teensy 4.1 Code

See [Google Folder Link](#) & Annotations in the code

8.2. R Code

See [Google Folder Link](#) & Annotations in the code

8.3. CAD Files for 3D Prints

See [Google Folder Link](#)

8.4. Diagrams & Photos

See [Google Folder Link](#)

8.5. Supplementary

- How to Build an Animal RFID reader with Arduino:
 - ▶ #263 How to build an Animal RFID reader with Arduino, ESP8266, or ES...
 - This helps you understand the reason why/how the components work together.
- How to solder: ▶ Soldering Tutorial for Beginners: Five Easy Steps
- How to install and run Arduino IDE:
 - ▶ How to Install Arduino Software IDE on Computer / Laptop
- How to install and run R Studio:
 - ▶ How to install R and RStudio on Windows 11 (Updated 2025)
- How to use Onshape: ▶ Onshape Tutorial for Absolute Beginners!
 - Onshape is a web-based 3D modeling platform, where you can make your 3D model for your antenna housing.
 - An alternative way of coiling your antenna could be using a rigid PVC pipe (circular) or a short hollow wooden bar (rectangular). You may need a router or table saw to cut down the original material.

